



Is AI the new corporate monitor? Evidence from excessive on-the-job consumption

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ABSTRACT

Excessive on-the-job consumption (EOJC) represents a manifestation of corporate agency problems that ultimately undermines firm value. As artificial intelligence (AI) applications permeate various aspects of corporate operations, this study examines AI's impact on EOJC. Analyzing 2788 Chinese manufacturing firms (2007–2022), we find AI implementation significantly reduces EOJC by dissolving hierarchical information barriers to enhance governance quality and curbing managerial discretion through algorithm-driven decision protocols. Further analysis reveals that AI exerts stronger governance effects in highly competitive industries by augmenting existing market discipline mechanisms, while synergizing with information-transparent firms to amplify monitoring efficacy. These findings reveal AI's potential for institutional transformation in corporate governance systems.

1. Introduction

Within modern corporate governance frameworks, excessive on-the-job consumption (EOJC) emerges as a quintessential manifestation of principal-agent problems, directly escalating agency costs (Gul et al., 2011) while systematically eroding organizational value through financial misconduct (Xu et al., 2014) and distorted resource allocation (Cai et al., 2011). Classical agency theory posits that unchecked managerial power enables the expropriation of positional privileges for private gain (Jensen and Meckling, 1976), a phenomenon exacerbated in firms with opaque information environments and deficient monitoring mechanisms (Yalamov and Belev, 2002). Of particular concern in high-tech industries is executives' strategic manipulation of technological trajectories and covert collusion with preferred suppliers to secure excess consumption opportunities. Such practices - where managerial perquisites transcend operational necessities to maximize self-interest (Gen and Wang, 2016)-jeopardize the cultivation of core competitive advantages. While prior studies have explored governance mechanisms spanning board oversight (Zhang and Pan, 2009), institutional investor monitoring (Li et al., 2012), and market-oriented reforms (Xin and Tan, 2009) to erect "firewalls" against covert rent-seeking, these conventional approaches now confront the transformative potential of Artificial

intelligence (AI). As algorithmic systems revolutionize corporate monitoring through real-time data analytics and autonomous execution, our central inquiry becomes: Can AI overcome traditional oversight limitations to fundamentally reshape the constraints on executive opportunism?

The pervasive integration of AI technologies into corporate ecosystems is fundamentally reconfiguring governance architectures. From a monitoring perspective, AI enables behavioral transparency through real-time data analytics (Gao and Liu, 2018), employs predictive algorithms for reputation risk mitigation (Hirsch, 2018), and facilitates decentralized authority structures via blockchain-enabled autonomous organizations (Murray et al., 2021). At the executive level, AI decision algorithms restrict managerial discretion over critical resources (Chen et al., 2020), while smart contracts' self-executing protocols minimize discretion-induced rent extraction (Duan et al., 2019). Most significantly, AI-powered audit systems overcome the inherent delays in conventional oversight (Issa et al., 2016), fundamentally altering the supervisor-supervisee dynamic. Nevertheless, extant literature exhibits fragmented understanding of AI's governance implications: Micro-level analyses predominantly treat AI as a technical instrument for decision automation (e.g., Duan et al., 2019), neglecting its potential as a "digital monitor" to constrain implicit agency costs. Meanwhile, macro-level

AbstractThis study examines how artificial intelligence mitigates executive excessive on-the-job consumption by enhancing corporate governance and constraining managerial discretion, with effects amplified in competitive and transparent firms. Findings underscore AI's institutional role in modern corporate governance.

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studies primarily address AI's impact on organizational structures (Qi and Xiao, 2020), leaving significant gaps regarding AI's interactions with executive behavior, particularly within emerging market institutional contexts.

This study transcends conventional “institution-behavior” paradigms by integrating AI into corporate governance theory, leveraging data from 2788 Chinese manufacturing listed firms to unravel AI's efficacy in mitigating EOJC. Our contributions are threefold: First, we identify novel constraint mechanisms through which AI, as an “impersonal monitor,” mitigates implicit agency costs via data transparency (reducing information filtering to enhance governance quality) and procedural consistency (limiting discretionary authority to prevent relationship-based opportunism). Second, we delineate the heterogeneous boundaries of technological governance, revealing AI's “supervisory efficacy augmentation” effect in high-competition markets and AI's “synergistic enhancement” effect in high-transparency firms, thereby advancing microfoundations for AI adaptability theory. These findings advance governance theory by elucidating the efficacy of AI across organizational contexts, offering a transformative framework for addressing principal-agent conflicts through algorithmic oversight systems that transcend traditional institutional boundaries.

2. Research design

2.1. Data

This study examines Chinese A-share listed manufacturing firms from 2007 to 2022, drawing data from three sources: the IRPDB Intellectual Property Database for AI patent information, CNINFO website for annual reports, and CSMAR and WIND databases for financial indicators. To ensure data integrity, we applied five screening criteria: excluding observations with missing key variables, removing firms under special treatment status (ST, ST*, PT), eliminating companies with asset-liability ratios outside the 0–1 range, excluding firms with net profit margins exceeding 1, and winsorizing main continuous variables at the 1 % and 99 % levels. The final sample comprises 2788 firms.

2.2. Empirical specifications

Dependent variable: Following Luo et al. (2011), we measure excessive on-the-job consumption (EOJC) as the difference between actual managerial consumption and predicted normal consumption based on economic factors. Detailed methodology is provided in Appendix A.

Explanatory variables: Our study employs a dual-dimension AI adoption index combining managerial attention and technical implementation. For the attention dimension, we analyze AI-related terminology (“artificial intelligence,” “machine learning,” “deep learning,” and other related terms) in MD&A sections of firms' annual reports. The technical dimension quantifies patents containing AI-related terms (“automatic,” “intelligent,” and similar AI-focused terminology) in their titles, applying a log transformation to (patent count + 1). The final index is constructed as the multiplicative interaction of these dimensions (Attention × Technical), ensuring our measure captures both strategic commitment and technological capability while mitigating single-source measurement biases.

Control variables: Throughout the observation period, notable disparities were evident across the sample in terms of State-Owned Enterprise (SOE) status, current ratio (cr), total asset growth rate (tagr), return on assets (roa), return on equity (roe), firm size, firm age (lnage), total debt ratio (tl), asset tangibility (tang), cash flow (cflow), and firm depreciation and amortization (Indepamo). These variations underscore the significant heterogeneity among the sampled firms. Table 1 presents the summary statistics for key variables.

Table 1

Summary Statistics.

VarName	Obs	Mean	SD	Min	Median	Max
EOJC	20,699	−0.000	0.022	−0.095	−0.003	0.147
AI	20,045	0.985	3.129	0.000	0.000	31.513
SOE	22,929	0.314	0.464	0.000	0.000	1.000
Cr	22,929	0.582	0.171	0.097	0.592	0.971
Tagr	22,928	0.210	0.381	−0.320	0.104	2.380
Roa	22,929	0.045	0.065	−0.247	0.043	0.287
Size	22,929	21.960	1.192	19.013	21.799	26.094
Lnage	22,551	1.906	0.948	0.000	2.079	3.332
Tl	22,929	0.398	0.196	0.054	0.388	0.898
Tang	22,910	0.375	0.155	0.016	0.362	0.797
cflow	22,925	0.052	0.070	−0.186	0.050	0.263
Indepamo	22,918	18.108	1.384	13.801	17.995	22.211

2.3. Models

To examine the relationship between AI implementation and EOJC, we estimate the following model:

$$EOJC_{it} = \alpha + \beta \text{firm_AI}_{it} + X_{it}\theta + \mu_t + \gamma_i + \delta_k + \varepsilon_{it} \quad (1)$$

where β is our coefficient of interest, with a negative value indicating that AI implementation constrains EOJC. X_{it} represents a vector of control variables. μ_t , γ_i , δ_k represent year, firm, and industry fixed effects, respectively, which control for unobserved time-invariant heterogeneity across these dimensions. ε_{it} is the idiosyncratic error term.

3. Results and analysis

3.1. Baseline results

Table 2 presents the baseline relationship between AI development and EOJC. Both measures are standardized with zero mean and unit

Table 2

Baseline Results.

VARIABLES	(1) EOJC	(2)
AI	−0.0235** (0.0090)	−0.0150** (0.0065)
SOE		0.3780* (0.1905)
cr		−0.2809*** (0.0937)
tagr		0.5834*** (0.0514)
roa		−0.0466 (0.2259)
size		−0.6391*** (0.0782)
lnage		0.2308*** (0.0436)
tl		0.0366 (0.2098)
tang		−0.2666* (0.1333)
cflow		0.4946*** (0.1150)
Indepamo		0.3053*** (0.0316)
Constant	0.0175*** (0.0001)	8.0810*** (1.3661)
Year FE	YES	YES
Firm FE	YES	YES
Industry FE	YES	YES
Observations	17,964	17,952
Adjusted R-squared	0.551	0.587

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors are clustered at the industry level in parentheses.

standard deviation. The AI coefficient remains negative and statistically significant at the 5 % level in both specifications (-0.0235 and -0.0150 , respectively). This indicates that a one standard deviation increase in AI development is associated with a reduction in EOJC of 0.0235 standard deviations without controls and 0.0150 standard deviations with full controls. These results provide empirical support for AI development in mitigating EOJC.

3.2. Robustness tests

To verify the robustness of our main findings, we conduct six alternative specifications. Table 3 presents these results. In Column (1), we replace our AI measure with the count of AI-related patents. Column (2) employs an alternative measurement of EOJC, with detailed calculation methodology provided in Appendix A. Column (3) incorporates city-level fixed effects to control for geographic heterogeneity. Column (4) excludes observations from 2020 to address potential COVID-19 pandemic disruptions. Column (5) adjusts standard error clustering from industry-level to firm-level. Finally, Column (6) employs a one-period lag of the AI variable to account for potential implementation delays. Across all specifications, the negative relationship between AI implementation and EOJC remains statistically significant, confirming that our findings are robust to alternative measurements, sample restrictions, and econometric approaches.

3.3. Endogeneity issues

First, to address dynamic endogeneity and potential simultaneity between AI adoption and EOJC, Column (1) of Table 4 presents the System GMM results. The AI coefficient remains negative and significant at the 1 % level (-0.0163), confirming AI's constraining effect on EOJC while addressing dynamic endogeneity. Model validity is confirmed by the Hansen test ($p = 0.619$) and Arellano-Bond tests, which show expected AR(1) autocorrelation ($p = 0.000$) without significant AR(2) autocorrelation ($p = 0.053$).

Second, to address self-selection bias, we implement the Heckman procedure. Column (2) reports the second-stage results with the AI coefficient remaining negative and significant (-0.0228 , $p < 0.01$). The significant negative lambda coefficient (-0.6564 , $p < 0.01$) in Column (3) indicates that unobservable factors increasing AI adoption likelihood are associated with lower EOJC, supporting our hypothesis. Both approaches consistently demonstrate that AI implementation significantly reduces EOJC after addressing endogeneity concerns.

Third, to address potential omitted variables bias, we implement a difference-in-differences (DID) strategy based on the staggered introduction of the National New Generation AI Innovation Development Pilot Zones. Column (4) shows that the AI coefficient remains

Table 3
Robustness Checks.

VARIABLES	(1) Replace IV	(2) Replace DV	(3) City FE	(4) Exclude COVID Year	(5) Change SE	(6) Lag Effect Test
AIpatent	-0.0321^{***} (0.0103)					
AI		-0.0233^{***} (0.0049)	-0.0148^{**} (0.0065)	-0.0161^{**} (0.0076)	-0.0150^{***} (0.0056)	
L.AI						-0.0177^{**} (0.0077)
Constant	8.0148^{***} (1.3799)	9.3331^{***} (1.2678)	8.1037^{***} (1.4078)	8.2108^{***} (1.3831)	8.0810^{***} (0.5651)	8.1386^{***} (1.4263)
Control	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES
City FE	NO	NO	YES	NO	NO	NO
Observations	17,952	17,952	17,951	16,307	17,952	17,194
Adjusted R-squared	0.587	0.657	0.589	0.588	0.587	0.596

Table 4
Endogeneity tests.

VARIABLES	(1) GMM EOJC	(2) Heckman /mills	(3)	(4) DID EOJC
AI	-0.0163^{***} (0.0058)	-0.0228^{***} (0.0071)		-0.0041^{***} (0.0007)
Lambda			-0.6564^{***} (0.0984)	
L.EOJC	0.5942^{***} (0.1352)			
Constant	-0.1188 (0.1109)	-0.1089^{**} (0.0539)		0.1663^{***} (0.0268)
Control	YES	YES	YES	
AR(1)	0.000			
AR(2)	0.053			
Hansen test P value	0.619			
Observations	15,894	20,438	20,438	20,360

significantly negative (-0.0041 , $p < 0.01$), supporting a causal interpretation. To validate the DID design, we conduct a parallel trends test (Fig. 1), kernel-based propensity score matching (Fig. 2), and a placebo test (Fig. 3). All robustness checks, reported in Appendix C, confirm the credibility of the DID estimates.

3.4. Mechanism analysis

AI constrains excessive on-the-job consumption by executives through two pathways: improving corporate governance and limiting executive power. Related calculations are presented in Appendix B.

As shown in Table 5, the mechanism tests reveal significant relationships between AI adoption and corporate governance measures. In Columns (1) and (2), AI shows a positive and statistically significant

Table 5
Mechanism test.

VARIABLES	(1) Management	(2)	(3) Power	(4)
AI	0.0056^{***} (0.0017)	0.0056^{***} (0.0015)	-0.0134^{***} (0.0040)	-0.0138^{***} (0.0039)
Constant	0.0629^{***} (0.0014)	2.5239^{***} (0.5813)	-0.0456^{***} (0.0040)	-2.3506^{***} (0.5950)
Control	NO	YES	NO	YES
Year FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES
Observations	12,894	12,880	19,198	18,863
Adjusted R-squared	0.834	0.843	0.760	0.762

Table 6
Results of heterogeneity tests.

VARIABLES	(1) HHI_low	(2) HHI_high	(3) Asymmetry_low	(4) Asymmetry_high
AI	−0.0290*** (0.0074)	−0.0043 (0.0099)	−0.0585** (0.0242)	−0.0140** (0.0054)
Constant	5.8111*** (1.4352)	11.4599*** (1.9544)	6.3658*** (1.7326)	8.5176*** (1.4372)
Control	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES
Observations	8774	8810	5744	12,000
Adjusted R-squared	0.628	0.646	0.607	0.594

coefficient (0.0056, $p < 0.01$) with the corporate governance index (Management), indicating that AI adoption is associated with improved governance quality. This positive relationship suggests that AI enhances monitoring capabilities and information transparency, which strengthens overall corporate governance mechanisms. In contrast, Columns (3) and (4) demonstrate a negative and statistically significant relationship between AI adoption and executive power (coefficients of -0.0134 and -0.0138 , both $p < 0.01$). This negative association confirms that by facilitating flatter organizational structures and network-based management, AI diminishes executives' ability to maintain information monopolies that traditionally served as a source of power. This structural reconfiguration limits the discretionary authority that executives might otherwise leverage for personal benefit.

3.5. Heterogeneity analysis

To examine how AI's effect on EOJC varies across different competitive environments, we employed the Herfindahl-Hirschman Index (HHI) to measure industry concentration. The HHI is calculated as the sum of squared market shares:

$$HHI = \sum_{i=1}^N \left(\frac{\text{Revenue}_i}{\text{Industry Total Revenue}} \right)^2 \quad (2)$$

Our results in Table 6 reveal a significant negative coefficient for AI in low HHI environments (Column 1: -0.0290 , $p < 0.01$), contrasting with the statistically insignificant coefficient in high HHI settings (Column 2: -0.0043). This indicates AI more effectively constrains executives' excessive consumption in highly competitive industries. In competitive markets, firms implement more rigorous AI-based

monitoring systems that strengthen internal controls against agency problems.

Information asymmetry represents a critical factor in corporate governance, as increased transparency reduces earnings management and improves monitoring effectiveness (Fang et al., 2016). We hypothesized that AI's effectiveness in constraining EOJC depends on information quality and transparency. Following Jin & Myers (2006), we measured information asymmetry using stock price synchronicity, with higher values indicating greater information asymmetry. Results in Table 6 demonstrate that AI implementation more effectively constrains EOJC in firms with low information asymmetry (Column 3: -0.0583 , $p < 0.01$) compared to firms with high information asymmetry (Column 4: -0.0129 , $p < 0.05$). These findings suggest AI primarily amplifies existing governance capabilities rather than transforming dysfunctional governance environments, highlighting the complementary relationship between technological monitoring systems and information transparency.

4. Conclusion

This study demonstrates that artificial intelligence serves as an effective corporate monitor, significantly reducing excessive on-the-job consumption through dual mechanisms: enhancing governance quality and restricting managerial discretion. Our empirical evidence reveals AI's governance effects are more pronounced in highly competitive industries and firms with greater information transparency, indicating that technological monitoring systems can transcend traditional institutional boundaries to alleviate principal-agent conflicts.

Appendix A

Dependent variable

Referring to the methods of Quan et al. (2010) and Luo et al. (2011), remuneration of directors, senior executives, and supervisors and amortization of intangible assets obviously do not belong to on-the-job consumption. The remaining amount after deducting these items from management expenses is the total amount of on-the-job consumption of managers (OJC). Excessive on-the-job consumption (EOJC) is equal to the difference between the manager's total on-the-job consumption and the expected normal OJC determined by economic factors. The specific calculation formula is as follows:

$$\frac{OJC_t}{\text{Asset}_{t-1}} = \beta_0 + \beta_1 \frac{1}{\text{Asset}_{t-1}} + \beta_2 \frac{\Delta \text{Sale}_t}{\text{Asset}_{t-1}} + \beta_3 \frac{PPE_t}{\text{Asset}_{t-1}} + \beta_4 \frac{\text{Inventory}_t}{\text{Asset}_{t-1}} + \beta_5 \text{LnEmployee}_t + \varepsilon \quad (A1)$$

Where OJC_t represents the total amount of OJC of managers, which is the amount remaining after deducting the remuneration of directors, senior executives, and supervisors and amortization of intangible assets and other items that are obviously not part of OJC from the administrative expenses. Asset_{t-1} is the total assets of the enterprise in the previous period. ΔSale_t is the change in main business income. PPE_t is the net fixed assets of the current period. Inventory_t is the total inventory of the current period, and LnEmployee_t is the natural logarithm of the number of employees of the

enterprise. We run regressions on the samples by year and industry, and the predicted value of the dependent variable indicates the total amount of normal on-duty consumption of managers. The difference between the total amount of OJC of managers that actually occurs and the expected total amount of OJC indicates the excessive on-the-job consumption of managers (EOJC).

Robustness test (Replace DV):

Following Zhai et al. (2015), we conducted robustness tests using EOJC2 as an alternative measure of excessive on-the-job consumption. The model proposed by Quan et al. (2010) may present two primary limitations: First, it fails to account for regional disparities, which tends to overestimate managerial OJC in developed regions such as eastern provinces while underestimating it in less developed areas. Second, the model does not adequately estimate reasonable components of management expenses at the discretion of executives, such as costs associated with board meetings, shareholder general meetings, and various functional committee meetings. To address these limitations, we propose the following revised model (2):

$$\frac{EOJC_{i,t}}{Asset_{i,t-1}} = \beta_0 + \beta_1 \frac{1}{Asset_{i,t-1}} + \beta_2 \frac{\Delta Sale_{i,t}}{Asset_{i,t-1}} + \beta_3 \frac{PPE_{i,t-1}}{Asset_{i,t-1}} + \beta_4 \frac{Inventory_{i,t-1}}{Asset_{i,t-1}} + \beta_5 \frac{ZGXC_{i,t}}{Asset_{i,t-1}} + \beta_6 Canhui_{i,t} + \beta_7 Meeting_{i,t} + \varepsilon_{i,t} \quad (A2)$$

Where the subscript i represents the i th company. The variables in the equation are the same as the measure of EOJC. ZGXC represents the sum of employee compensation and employee welfare, and this variable is the product of the total number of employees and the average compensation of each employee. Obviously, the average compensation per employee is significantly related to region, and the variable ZGXC can better control regional differences. The variable Canhui = $\ln(1 + \text{the number of participants of shareholders' representative meetings and extraordinary general meetings})$, which reflects the number of participants of shareholders' representative meetings and extraordinary general meetings; Meeting = $\ln(1 + \text{the number of meetings of the seven categories})$. The seven categories of meetings include the board of directors, the board of directors held in the form of communication, the meeting of the compensation committee, the meeting of the audit committee, the meeting of the nomination committee, the meeting of the strategy committee, and the meeting of the Board of supervisors. Obviously, the number of participants, the number of meetings, etc., are listed in the administrative expenses, which are also reasonable expenses in the administrative expenses. Similar to the measurement of EOJC, this study conducts annual and industry-specific regressions on all A-share company samples in the model, and the residual of the regression is the variable EOJC2 used in this study to quantify excess on-the-job consumption.

Appendix B

Mechanism variables

1. Corporate governance index

Following Gu and Zhou (2017), we construct a comprehensive corporate governance index using principal component analysis across three dimensions: supervision, incentives, and decision-making. For the incentive mechanism, we employ executive compensation (Mana_Pay) and executive shareholding ratio (Mana_Share). To capture the board's supervisory function, we include the proportion of independent directors (Otratio) and board size (Board). The supervisory role of ownership structure is represented by institutional shareholding ratio (Inst_Share) and the degree of ownership balance (Share_Balance), calculated as the sum of shareholding ratios from the second to fifth largest shareholders divided by the controlling shareholder's ratio. Finally, CEO duality (Dual) represents managerial decision-making power. Based on these seven indicators, we apply principal component analysis and adopt the first principal component as our comprehensive index reflecting corporate governance quality.

2. Executive Power

Following Quan and Wu (2010) and Zhou et al. (2016), we construct a comprehensive managerial power index (Power) through principal component analysis of five indicators: First, CEO duality is measured as a binary variable equal to 1 when the CEO concurrently serves as board chairman and 0 otherwise. CEO duality typically indicates greater managerial power. Second, board size is measured as the total number of directors in the given year, with larger boards generally associated with stronger managerial power. Third, the proportion of inside directors is calculated as the ratio of non-independent directors to total board size. Higher proportions of inside directors suggest greater managerial influence. Fourth, ownership dispersion is measured as the ratio of shareholdings by the 2nd through 10th largest shareholders to the largest shareholder's stake. Greater dispersion typically enables stronger managerial power. Fifth, managerial ownership is measured as the percentage of shares held by executives, with higher ownership typically reinforcing managerial power. The composite managerial power index increases with the concentration of executive authority, with higher values indicating stronger managerial power.

Appendix C

To ensure the reliability of the difference-in-differences (DID) estimation, we perform the following robustness checks:

Parallel Trends Test (Fig. 1): We employ an event-study specification to examine whether the treatment and control groups exhibited parallel trends in the outcome variable prior to the policy intervention. The results confirm no significant pre-treatment differences, satisfying the key identifying assumption of DID.

Kernel Matching Estimation (Fig. 2): To improve covariate balance and reduce potential selection bias, we apply kernel-based propensity score matching before implementing the DID regression. The matched-sample results remain consistent with our main findings.

Placebo Test (Fig. 3): We conduct a falsification test by randomly assigning pseudo-treatment years in the pre-policy period. The resulting placebo coefficients are statistically insignificant, indicating that the observed treatment effects are unlikely to be spurious.

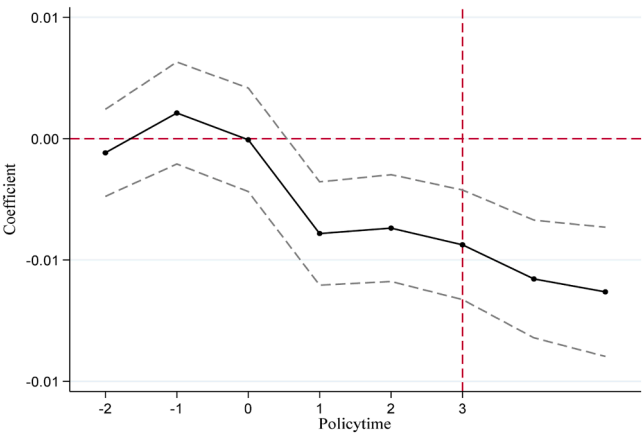


Fig. 1. Parallel Trends Test.

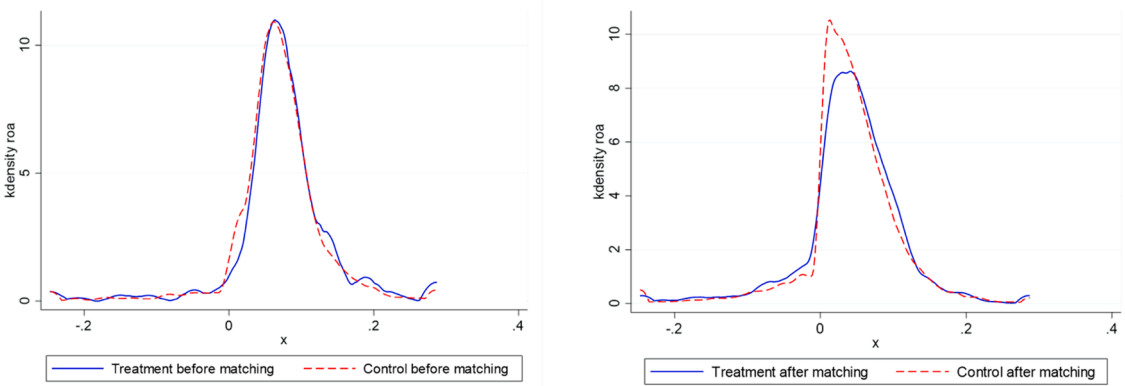


Fig. 2. Kernel Matching Estimation.

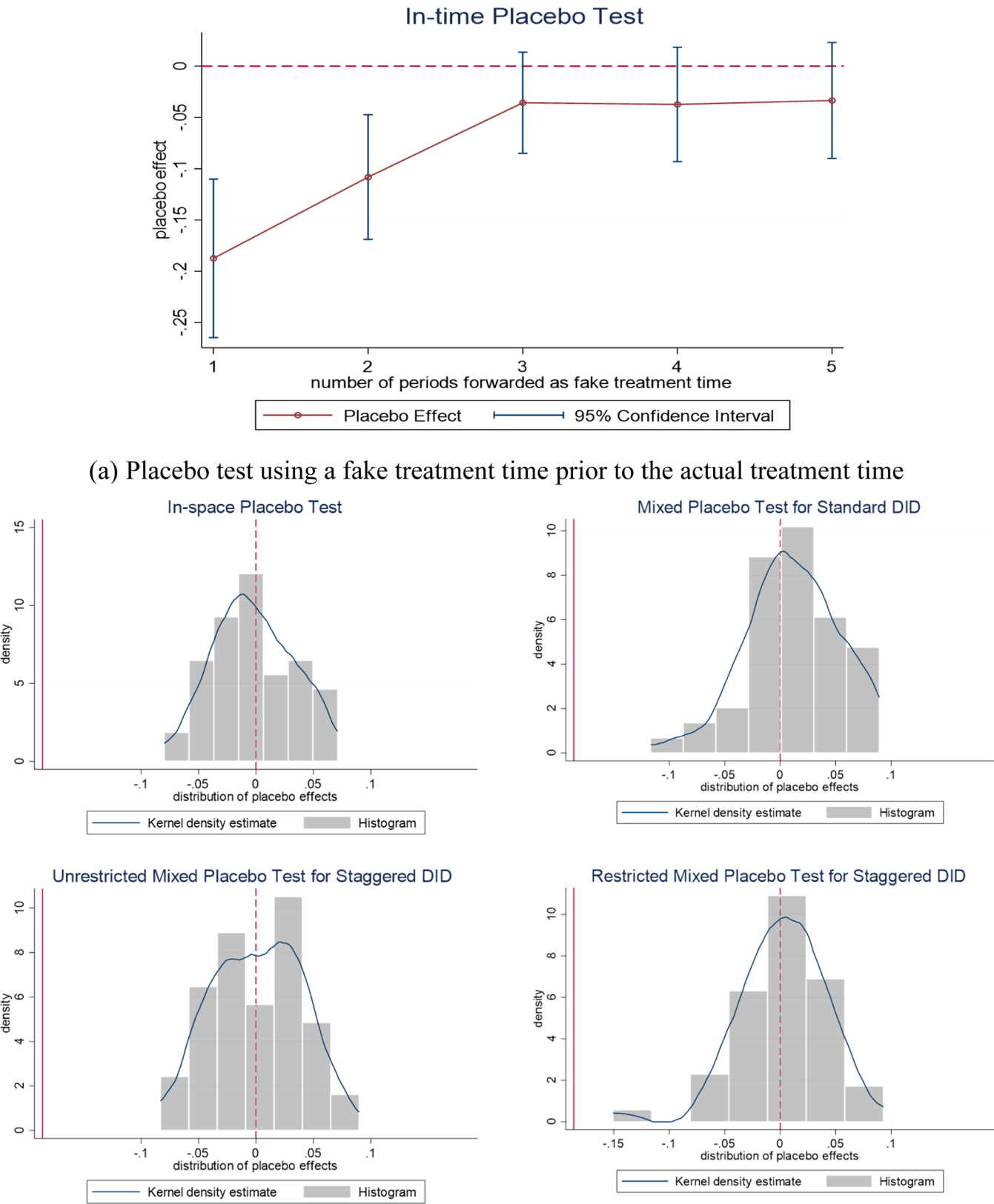


Fig. 3. Placebo Test.

Data availability

The authors do not have permission to share data.

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